Skills Summary

Average Value, Accumulated Change, and Motion

Use this reference while completing your worksheet or when studying.

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1. Average value of a function

Rule: the average value of f on $[a, b]$ is $f_{\text{avg}} = \frac{1}{b-a}\int_a^b f(x)\,dx$. Ask "what is the typical height?" — the integral gives area, and dividing by the width $b-a$ turns that area back into a height. The answer carries the units of f, never the units of the integral. Warning sign that you have forgotten the division: an answer that grows when the interval gets longer.

Example: Find the average value of $f(x) = 6x$ on $[1, 4]$.
Step 1. The question asks for a typical height rather than a total, so this is the average-value formula: integrate, then divide by the interval width.
Step 2. Width $= 4 - 1 = 3$, so compute $\int_1^4 \frac{6x}{3}\,dx = \int_1^4 2x\,dx = [x^2]_1^4 = 16 - 1$. $f_{\text{avg}} =$ **15**

Try it: Find the average value of $f(x) = 4x$ on $[0, 5]$.

check: **10**

2. Accumulated change from a rate

Rule: if $r(t)$ is a rate of change, then $\int_a^b r(t)\,dt$ is the *change* in the quantity between $t = a$ and $t = b$, and the amount at the end is $Q(b) = Q(a) + \int_a^b r(t)\,dt$. The integral alone is never the final amount unless the starting amount is zero. When two rates act at once, integrate the net rate: in minus out.

Example: Water enters a tank at $r(t) = 2t + 1$ litres per minute. How much enters during the first 4 minutes?
Step 1. The given function is a rate and the question asks for an amount, so integrate the rate over the time interval — no division here, unlike an average value.
Step 2. $\int_0^4 (2t + 1)\,dt = [t^2 + t]_0^4 = 16 + 4$. Water added $=$ **20** litres

Try it: How much enters in the first 4 minutes at $r(t) = 3t + 2$ litres per minute?

check: **32**

3. Net displacement from velocity

Rule: for a particle with velocity $v(t)$, the net displacement on $[a, b]$ is $\int_a^b v(t)\,dt$, and the position is $s(b) = s(a) + \int_a^b v(t)\,dt$. The integral is signed: an interval where $v < 0$ subtracts. A negative answer is not an error — it means the particle finished behind where it started.

Example: A particle has $v(t) = 2t - 5$ metres per second on $[0, 4]$. Find its net displacement.
Step 1. "Displacement" asks where the particle ended relative to where it started, so integrate the velocity itself and keep the signs.
Step 2. $\int_0^4 (2t - 5)\,dt = [t^2 - 5t]_0^4 = 16 - 20$. Displacement $= \mathbf{-4}$ metres

Try it: Find the net displacement for $v(t) = 2t - 8$ on $[0, 5]$.

check: $\mathbf{-15}$

4. Total distance travelled

Rule: total distance on $[a, b]$ is $\int_a^b |v(t)|\,dt$. Absolute values are not integrated directly: solve $v(t) = 0$ first, split the interval at every sign change inside it, integrate each piece, then add the absolute values of the pieces. Total distance is always $\geq$ the size of the displacement, and the two agree exactly when v never changes sign.

Example: Find the total distance for $v(t) = t - 3$ on $[0, 5]$.
Step 1. "Total distance" means integrate speed, so find the sign change first: $v = 0$ at $t = 3$, which is inside the interval, so the trip splits there.
Step 2. $\int_0^3 (t - 3)\,dt = -4.5$ and $\int_3^5 (t - 3)\,dt = 2$; add the sizes: $4.5 + 2$. Total distance $= \mathbf{6.5}$ units

Try it: Find the total distance for $v(t) = t - 2$ on $[0, 6]$.

check: **10**

(!) Watch out: Average *velocity* is displacement over elapsed time; average *speed* is total distance over elapsed time. They are equal only when the velocity never changes sign. Read the question for the word "total" — it is the single word that decides whether an absolute value belongs in your integral.